

Jenkins Installation Using Shell Script

Assignment

Create a Shell script to install Jenkins on a Linux machine. The Jenkins WAR version must be passed by the user as a command-line argument.

1. Prerequisites

- Linux machine
- Root or sudo access
- Internet connectivity
- Java 21
- wget

2. Usage

Make the script executable:

```
chmod +x install-jenkins.sh
```

Run the script by passing the Jenkins version:

```
sudo ./install-jenkins.sh 2.568.3
```

3. What the Script Does

1. Accepts the Jenkins WAR version as a command-line argument.
2. Validates that a Jenkins version was provided.
3. Checks that the script is run with root privileges.
4. Installs Java and wget.
5. Creates the Jenkins installation and JENKINS_HOME directories.
6. Downloads the requested Jenkins WAR file.
7. Starts Jenkins on port 8080.
8. Displays the Jenkins process ID, URL, log location, and initial administrator password.

4. Jenkins Installation Script

```
#!/bin/bash

set -e

JENKINS_VERSION="$1"

BASE_DIR="/mnt"
JENKINS_DIR="${BASE_DIR}/jenkins"
JENKINS_HOME="${JENKINS_DIR}/jenkins_home"

JENKINS_WAR="jenkins.war"
JENKINS_URL="https://get.jenkins.io/war-stable/${JENKINS_VERSION}/jenkins.war"
JENKINS_PORT="8080"

if [ -z "${JENKINS_VERSION}" ]; then
    echo "ERROR: Jenkins version is required."
```

```

        echo "Usage: sudo ./install-jenkins.sh <jenkins-version>"
        echo "Example: sudo ./install-jenkins.sh 2.568.3"
        exit 1
    fi

    if [ "$EUID" -ne 0 ]; then
        echo "ERROR: Please run this script as root or with sudo."
        exit 1
    fi

    echo "=====
    echo " Jenkins Installation"
    echo "=====
    echo "Jenkins Version : ${JENKINS_VERSION}"
    echo "Installation Dir: ${JENKINS_DIR}"
    echo "Jenkins Home      : ${JENKINS_HOME}"
    echo "Jenkins Port       : ${JENKINS_PORT}"
    echo "=====

    echo "Installing Java and wget..."
    yum install -y java-21-amazon-corretto.x86_64 wget

    echo "Checking Java installation..."
    java -version

    echo "Creating Jenkins directories..."
    mkdir -p "${JENKINS_DIR}"
    mkdir -p "${JENKINS_HOME}"

    echo "Downloading Jenkins WAR ${JENKINS_VERSION}..."
    cd "${JENKINS_DIR}"
    wget -O "${JENKINS_WAR}" "${JENKINS_URL}"

    if [ ! -f "${JENKINS_DIR}/${JENKINS_WAR}" ]; then
        echo "ERROR: Jenkins WAR download failed."
        exit 1
    fi

    export JENKINS_HOME="${JENKINS_HOME}"

    echo "Starting Jenkins on port ${JENKINS_PORT}..."
    cd "${JENKINS_DIR}"

    nohup java -jar "${JENKINS_WAR}" \
        --httpPort="${JENKINS_PORT}" \
        > "${JENKINS_DIR}/jenkins.log" 2>&1 &

    JENKINS_PID=$!

    echo "Waiting for Jenkins to start..."
    sleep 10

    if ps -p "${JENKINS_PID}" > /dev/null 2>&1; then
        echo "=====
        echo " Jenkins Started Successfully"
        echo "=====
        echo "Jenkins Version : ${JENKINS_VERSION}"
        echo "Jenkins PID       : ${JENKINS_PID}"
        echo "Jenkins Home      : ${JENKINS_HOME}"
        echo "Jenkins Log       : ${JENKINS_DIR}/jenkins.log"
        echo "Jenkins URL       : http://<SERVER-IP>:${JENKINS_PORT}"
        echo "=====

```

```

else
    echo "ERROR: Jenkins failed to start."
    echo "Check: cat ${JENKINS_DIR}/jenkins.log"
    exit 1
fi

echo ""
echo "Initial Jenkins Admin Password:"
if [ -f "${JENKINS_HOME}/secrets/initialAdminPassword" ]; then
    cat "${JENKINS_HOME}/secrets/initialAdminPassword"
else
    echo "Password file is not available yet."
    echo "Check: cat ${JENKINS_HOME}/secrets/initialAdminPassword"
fi

echo ""
echo "View logs: tail -f ${JENKINS_DIR}/jenkins.log"
echo "Check process: ps -ef | grep jenkins"
echo "Check port: ss -lntp | grep ${JENKINS_PORT}"

```

5. Example Execution

```
sudo ./install-jenkins.sh 2.568.3
```

6. Jenkins Access

Open `http://<SERVER-IP>:8080` in a browser. Replace `<SERVER-IP>` with the Linux machine IP address.

7. Initial Administrator Password

```
/mnt/jenkins/jenkins_home/secrets/initialAdminPassword
```

```
Command: cat /mnt/jenkins/jenkins_home/secrets/initialAdminPassword
```

8. Useful Commands

```
tail -f /mnt/jenkins/jenkins.log
```

```
ps -ef | grep jenkins
```

```
ss -lntp | grep 8080
```

9. GitHub Repository Structure

```

Jenkins-assignment/
├── jenkins-installation/
│   ├── install-jenkins.sh
│   ├── README.md
│   ├── .gitignore
│   └── Jenkins-Installation-Assignment.pdf

```

10. Conclusion

The script provides a reusable method to install Jenkins on Linux while allowing the Jenkins WAR version to be supplied dynamically during execution.